Schooling and college inside the adjusted CH index

Controls: CH specification—County-adjusted CH; Schooling and college; Geography + schooling and college.

1. Schooling and college. Let q_c, n_c and a_c denote county GDP, employment and land area, with density $d_c = n_c/a_c$. Let S_c be average years of schooling and C_c the college share between zero and one. Define $h_c = \beta S_c$, using the fixed micro-based schooling return $\beta = 0.102623$. The college coefficient is fixed at $\kappa = 0.75$. For MSA or state g, let $N_g = \sum_{c \in g} n_c$. Selecting Schooling and college gives

$$D_g^{\mathrm{HC}}(\theta) = \frac{1}{N_g} \sum_{c \in g} \exp(h_c + \kappa C_c)\, n_c^{\theta} a_c^{1-\theta}.$$

Each county's schooling and college share multiply that county's contribution before aggregation. State analysis uses the same historical FE-defined county groups as standard CH. This specification needs no county regression or geography coefficients, so missing geography inputs do not remove regions.

2. Geography + schooling and college. This option adds county geography and transport effects γ. The county regression uses $\log(q_c/n_c) - h_c - \kappa C_c$ as the outcome, with log density, selected geography variables X_c, modern state fixed effects and optional log county area as regressors. Under IV, historical population in levels instruments log density. The combined index is

$$D_g^{\mathrm{geo+HC}}(\theta) = \frac{1}{N_g} \sum_{c \in g} \exp(X_c' \widehat{\gamma} + h_c + \kappa C_c) n_c^{\theta} a_c^{1-\theta}.$$

The county density coefficient, state fixed effects and optional area coefficient do not enter this multiplier. Both indices use actual employment in density and in the denominator. Schooling is not multiplied by θ.

3. The regional outcome. Both specifications relate log GDP per employment to the selected log index, estimating $\rho = \theta - 1$. For all industries the outcome is $\log(Q_g/N_g)$, where Q_g is total GDP; an MSA sector choice uses that sector's GDP and employment. The regional outcome receives no further schooling or college correction. County human capital remains inside the sum because it differs across counties and multiplies their density contributions.

4. Sample and uncertainty. The schooling return and college coefficient are held fixed in estimation and inference. With geography, standard errors also account for estimated geography effects and shared county–regional observations. Every constituent county enters the index. Missing schooling, college share or another required input in a positive-employment county excludes the whole region. The geography specification additionally requires the selected geography inputs and an identified county regression. Compare observation counts when changing the adjustment.